Minor Project – III (MCA-269): Problem Analysis

Project	AI Tutor: An Evidence-Based Adaptive Learning System with Deliberate Scaffolding Withdrawal
Programme	Master of Computer Applications, Semester III, Batch 2025–2027
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Guide	_____
Date	24 September 2026 (first draft 7 September 2026)
Submission	Phase 1: Synopsis, Problem Analysis, Literature Survey, SRS

1. Purpose of this Document

The synopsis describes the problem briefly and then moves on to the solution. This document looks at the problem in more detail. It defines the problem, traces each weakness to its cause, lists who is affected, sets the objectives the system has to meet, fixes the scope, and checks feasibility before we start building. The SRS then turns these objectives into numbered requirements. Everything here is based on the synopsis or the literature survey.

2. Problem Definition

Students who learn technical subjects on their own have no tool that measures or improves their ability to build things without help. Video courses, coding websites, flashcard apps, general AI assistants and AI tutors all reward consumption: videos watched, lessons finished, streaks kept. None of them measure what the student can write alone.

When AI help is given without limits, the situation gets worse. In a controlled trial with nearly 1,000 students, students who had unrestricted AI help scored 17% lower afterwards than students who had no AI help at all (Bastani et al., 2025).

So the problem is not one missing feature in one product. Across the whole category there is no system that tracks reading and writing ability separately, updates its view of the student only from code that was actually run, and reduces its own help on a planned schedule. Every module in the SRS traces back to this problem.

3. Study of the Existing System

Here the "existing system" is not an old application that we are replacing. It is the way a self-learning student works today, using whatever tools they pick up at each step. Nobody designed this workflow and no single company is responsible for its problems.

Step	Who	What happens	Tools used today	Result
1	Student	Chooses a skill to learn	Job posting, interest, syllabus	–
2	Student	Picks a course	YouTube, Udemy, Coursera, docs	A course link, no record of the student
3	Student	Watches videos in order	Video player, playlist	Watch percentage
4	Student	Types along with the instructor	Editor next to the video	A copy of the instructor's code
5	Student	Marks lessons done	Checkbox	Completion percentage
6	Student	Tries to work alone	Empty editor, no support	Often nothing
7	Student	Gets stuck	–	–
8	Student	Asks an AI assistant	ChatGPT, Copilot, Claude	Working code, given on request
9	Student	Moves on	–	No lasting skill
10	Student	Returns weeks later	–	Has forgotten most of it; no revision was planned

Where it goes wrong

- Steps 3 to 5 train recognition, and the platform records this as progress.
- Step 6 is the first time the student has to build something. It comes as an empty file with no support in between, which is where students most often give up.
- Step 8 removes the immediate block, but it also removes the effort of recalling that would have produced learning. It also makes the student more confident that they understood (Roediger & Karpicke, 2006; Literature Survey §2.2).
- Step 10 cannot be prevented by anything in the earlier steps, because nothing schedules revision.

No step in this process keeps a lasting, evidence-based record of what the student can and cannot do. That missing record is the common cause behind every problem in Section 4.

4. Root-Cause Analysis

No.	Problem (synopsis §2.2)	Cause	Evidence	What happens if it is not fixed
D1	Completion is measured instead of ability	Products store clicks and views, not evidence of what the student built	Literature Survey §3.2: MOOC completion of 3.13%, and no provider measures ability after completion	Nobody can tell a capable student apart from one who just watched everything

No.	Problem (synopsis §2.2)	Cause	Evidence	What happens if it is not fixed
D2	Recognition is trained, recall is not	Videos and read-along formats only re-expose the student; this builds confidence, not retention	Roediger & Karpicke (2006): 61% vs 40% retention, and the re-reading group expected to do better	Students feel ready and are not; they find out only when they have to build something
D3	Courses are linear, students are not	No system stores a mastery score per skill, so none can choose a personal next step	My own test: 10/11 on reading code, 0/4 on writing it	Time goes into known topics while the real gap stays
D4	No system keeps a learner model	Same cause as D3, seen as a missing component	Corbett & Anderson (1994): Bayesian Knowledge Tracing exists but no consumer product we looked at uses it	Every "next step" is the student guessing
D5	AI assistants make it worse	They are tuned to be as helpful as possible in each reply, which removes the effort that produces learning	Bastani et al. (2025): 17% drop with unrestricted access; hint-only access avoided the drop	The newest and most used tool in the workflow harms learning unless limited
D6	Unrevised material is forgotten	No product tracks forgetting for skills (Anki and Quizlet do it, but for facts)	Cepeda et al. (2006): the revision gap should grow with how long you want to remember; no skill-building product we surveyed does this	Skills measured once are not skills kept; relearning keeps happening
D7	The empty file ends the session	No product adjusts task difficulty to the student's level; support is either always on (coding sites) or absent (practice sites)	Sweller & Cooper (1985) and the expertise reversal effect: worked examples help beginners but hurt students who are nearly experts	Students give up instead of progressing slowly

All of these come back to the same gap from Section 3: there is no lasting, evidence-based model of the individual student. That is why the main component of our system is the learner model (M4), and not a content library.

5. Stakeholders

Stakeholder	Role	What they want	Current difficulty	What they need from us
Student	Learns the skill	To be able to work independently, quickly	Cannot tell real progress from completion numbers; gives up at the empty file	An honest mastery score, tasks at the right level, and help that reduces over time

Stakeholder	Role	What they want	Current difficulty	What they need from us
Curriculum author	Writes and maintains the skill graph and sample exercises	A correct prerequisite structure that is easy to maintain	No tool keeps graph writing separate from content generation	An editing screen that checks for cycles, with a bank of hand-written exercises as backup
Administrator	Runs the platform	Cost control, accurate predictions, uptime	No way to see whether the mastery predictions are correct	Accuracy dashboards, logs of failed exercise generation, cost and usage figures
Institution or employer (buyer)	Pays for training and answers for results	Proof that training built ability	Only gets completion numbers	A skill profile for each student, backed by evidence (synopsis §7.2)
Evaluation panel	Assesses the project	A working system, sound design decisions, use of semester coursework	–	A system that every team member can explain and that can be demonstrated live

The student and the buyer want different things from the same data: the student wants to know what to do next, and the buyer wants proof of ability. The same evidence-based mastery model serves both, which is why this document and the SRS are built around it.

6. Objectives

Each objective addresses one or more of the problems in Section 4.

- **O1.** Keep two mastery scores per skill for each student (reading and writing), updated only from evidence produced by running code. *(D1, D2, D3, D4)*
- **O2.** Choose the next step with fixed rules, using the prerequisite graph and current mastery, not a fixed course order. *(D3, D4)*
- **O3.** Limit AI help in the system's design itself, through a hint level stored on the server and an output filter, not only through prompt instructions. Help should be available without replacing the student's own work. *(D5)*
- **O4.** Calculate how mastery fades over time and schedule revision before it falls below a threshold. *(D6)*
- **O5.** Offer five difficulty levels chosen from the writing score, so the student only reaches an empty file when they are ready for it. *(D7)*
- **O6.** Reduce support on a schedule linked to the student's plan, and not only when the student asks. Because of the expertise reversal effect (Literature Survey §2.5), this is needed for correctness. *(D5, D7)*

- **O7.** Log every time the system had to reveal a full answer as a failure, and use this to adjust difficulty. This keeps the mastery model honest about how much help was really given. *(supports O1, O5)*

7. Scope

7.1 In scope for version 1

- One subject: Data Structures and Algorithms. Topic coverage is checked against the roadmap.sh DSA roadmap. Prerequisite links, topic descriptions and exercises are written by us (synopsis §3.2).
- Text, code and generated exercises, each checked against a reference solution before use.
- The full learner model: separate reading and writing scores, exponential decay, and rule-based choice of the next step.
- The five-level exercise ladder and the hint engine (hint ladder, output filter, logging of answer reveals).
- Isolated code execution and grading in a sandbox.
- Basic curriculum editing and admin screens, enough for our team to run a pilot.

7.2 Out of scope (synopsis Appendix)

- Video content.
- More than one subject in version 1.
- Mobile apps.
- Social features, leaderboards and streaks.
- Certificates. The mastery graph is the record of skill.
- Payments, until the institutional phase.
- Training our own language models.
- Building our own code sandbox. We will use a managed service.

Anything added to the scope later has to replace something already in it (synopsis §3.2). We added this rule to avoid the scope creep that both the department guidelines and the literature warn about.

8. Feasibility Study

8.1 Technical feasibility

Every part of the stack is already mature and used in production: Next.js/React with TypeScript for the frontend, Monaco for the in-browser editor, Node/Express for the backend, PostgreSQL with pgvector for relational and vector data, Redis with BullMQ for sessions and background jobs, and a managed sandbox service (synopsis §5.2). The new work is not in the infrastructure. It is in the mastery maths (decay, weighting of evidence, scoring of the next step) and in the pipeline that

generates and checks exercises. Both are part of version 1 and are owned by one team member, with a clear API boundary to the rest of the system (synopsis §4.1).

Result: feasible. The exercise generation and checking pipeline (M6) carries the most technical risk. It is covered in the risk table below.

8.2 Operational feasibility

One hand-written DSA graph, with its topics checked against a well-known public roadmap, can be maintained by one or two curriculum authors, which matches our team. Admin work (checking accuracy and cost) is mostly dashboards and logs and does not need a separate operations team.

Result: feasible with our current team size and division of work.

8.3 Economic feasibility

During the pilot, the main running cost is paid API usage (language model calls and sandbox run time), not fixed servers. We use a small model for the many hint and grading requests and a larger model only for generating exercises, to keep this cost under control (synopsis §5.2). The validation pilot in synopsis §7.5 needs no upfront spending.

Result: feasible on a student-level API budget. We should fix a cost limit before Phase 2, once we have measured real token use per session.

8.4 Schedule feasibility

This document and the SRS are part of the Phase 1 submission on 24 September 2026. The remaining dates are system design and partial implementation on 7 October 2026, implementation, testing and draft report on 23 October 2026, and the final report and viva after that (synopsis §6.4). Nothing in this analysis changes those dates.

Result: feasible, as long as risks R5 and R7 in Section 9 are handled early.

8.5 Legal feasibility

The system stores learning data: attempts, scores and timestamps. It does not store health, financial or other specially protected data. Normal data minimisation and access control, which the SRS security requirements already ask for, are enough for a pilot. There are no extra legal obligations beyond normal handling of education data.

Result: feasible. Consent and data handling should still be written up in the final report.

9. Risk Analysis

ID	Risk	Type	Likelihood	Impact	How we handle it
R1	A generated exercise cannot be solved or its tests are wrong	Technical	Medium	High	Run the reference solution against the generated tests before use (M6); use a hand-written exercise if the check fails
R2	Code escapes the sandbox or uses up resources	Security	Low	High	Managed sandbox with no network and resource limits (M7); building our own sandbox is out of scope
R3	A new student has no evidence yet (cold start)	Product	High	Medium	The diagnostic test (M3) sets the starting reading and writing scores before the first step
R4	Examiners see one subject as too narrow	Academic	Medium	Medium	The literature survey and the methodology (synopsis §6.3) explain why we chose depth over breadth
R5	A student other than the author does not finish a multi-session project before submission	Schedule / Commercial	Medium	High	Already listed in synopsis §7.5; plan time for it well before Phase 3
R6	Language model costs exceed the pilot budget	Economic	Low to Medium	Medium	Use the large model only for generation (§8.3); measure real cost per session early
R7	Too much of the core work (learner model, recommender, generation, sandbox, hint engine) sits with one team member	Schedule	Medium	High	Agree on API contracts between modules early so frontend and UI work is not held up by the backend
R8	A student finds a way around the hint ladder or the filter	Product	Medium	Low	Two checks (server-side hint level and output filter, synopsis §3.6), and answer reveals are logged, not just blocked
R9	The pilot produces too little data to check the model's accuracy	Product	Medium	Medium	Held-back test exercises (M13) start collecting data from the very first pilot student

10. Conclusion

None of the products we surveyed keeps a lasting, evidence-based model of the individual student. The newest tool in the workflow, unrestricted generative AI, has been shown to make results worse when no such model exists. The feasibility study did not find anything that blocks the project on technical, operational, economic, schedule or legal grounds. The main risks are the exercise generation pipeline (R1), the workload on one team member (R7) and the validation check (R5), and all three can be tracked. Based on this and the literature survey, we move on to the SRS, which turns the objectives in Section 6 and the modules in the synopsis into numbered requirements.